Red Drum (*Sciaenops ocellatus*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.8	2.78	
Complexity in Reproductive Strategy	3.28	2.6	
Dispersal of Early Life Stages	2.44	2.42	
Egg and Larval Survival and Settlement Requirements	2.6	2.7	
Historical Variability in Temperature	1.64	2.6	
Juvenile Habitat Requirements	2.36	2.8	
Other Stressors	2.92	2.66	
Population Growth Rate	2.96	2.5	
Prey Specificity	1.44	2.5	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.88	2.7	
Spawning Cycle	2.48	2.8	
Stock Size/Status	2.56	2.1	
Total Sensitivity | Certainty	**Moderate (2) | 0.51**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.29	0.97	
Bottom Oxygen	2.29	0.85	
Bottom Salinity	2.6	0.86	
Bottom Aragonite Solubility	1.34	0.80	
Net primary production	1.06	0.17	
Downward Flux of Particulate Organic Carbon	1.15	0.36	
Total Exposure	**Moderate (1.74)**	**0.89**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (3.49)**	**1.77**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Red Drum. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate	Moderate	None
Exposure	Very High	Moderate	Decrease
Vulnerability	High	Low	Decrease
Directional Effect	Positive	Neutral	Decrease

Stocks

Red Drum is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

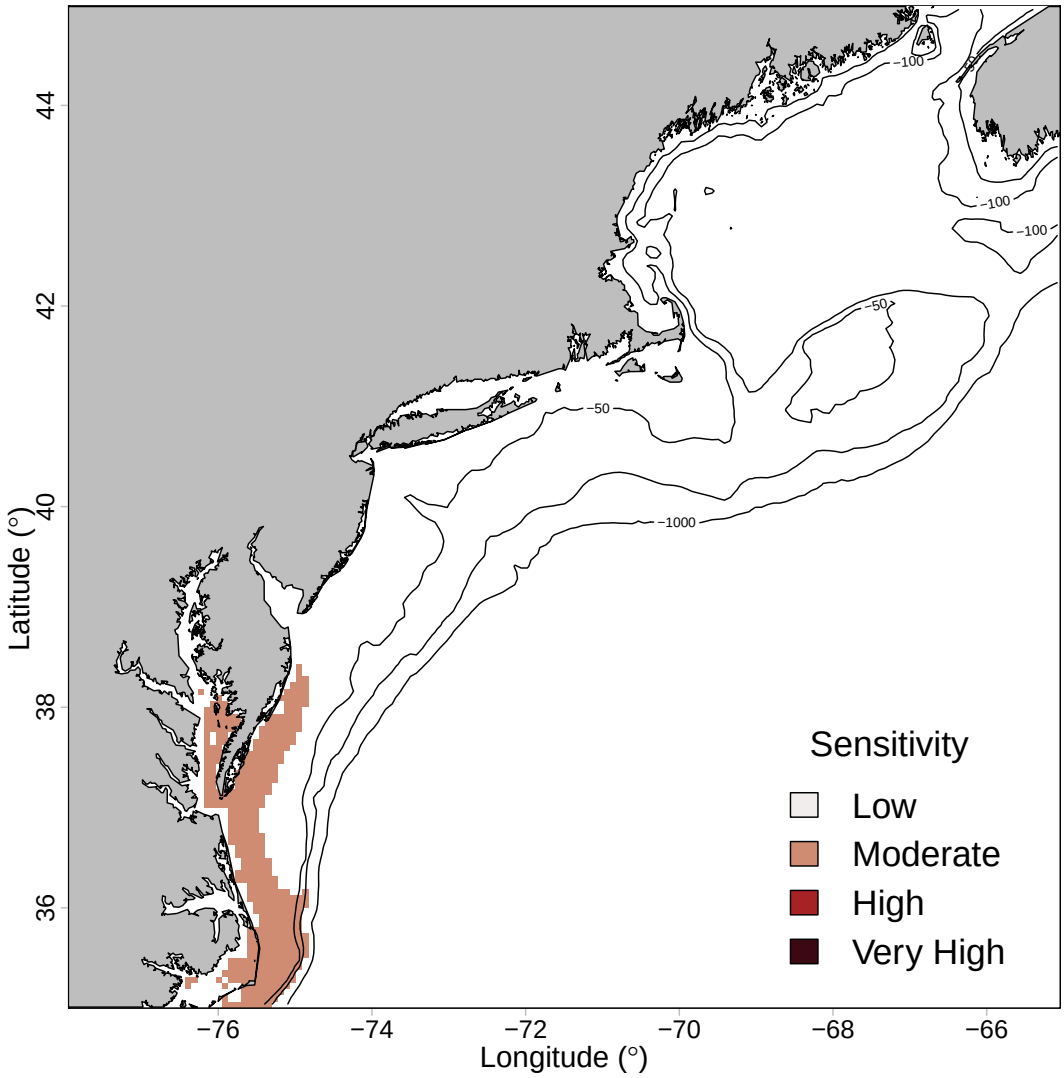


Figure 2: Red Drum Sensitivity on the Northeast Shelf

Red Drum received a **Moderate** sensitivity score with 50% certainty. This was driven by five attributes with scores over 2.5: Complexity in Reproductive Strategy (3.3), Population Growth Rate (3), Other Stressors (2.9), Egg and Larval Survival and Settlement Requirements (2.6), and Stock Size (2.6) (Figure 1).

Red Drum utilize spawning aggregations and spawn in estuaries, which are known vulnerable habitats (Guyant et al. 2022). Males also use sounds to attract females to spawning locations (Looby et al. 2022) (Montie et al. 2016). They mature at 3-4 years of age, and can live up to 62 years, making them a very long-lived species (Atlantic States Marine Fisheries Commission 2024). Larvae may rely on currents for transport into nursery habitats, which include seagrass beds.

Seagrass beds may contain an olfactory cue for settlement, and are highly vulnerable habitats in the Northeast US (Atlantic States Marine Fisheries Commission 2024) (Comyns et al. 1989) (Guyant et al. 2022). Adults are also dependent on shellfish reef habitats and estuarine water column habitats that are highly vulnerable (Guyant et al. 2022). Invasive species may also displace the native plant and prey species that red drum rely on (Atlantic States Marine Fisheries Commission 2024).

While the stock within the Northeast US is not overfished and overfishing is not occurring, models for the northern stock had high uncertainty and instability and could not be used to determine stock status in the most recent assessment (Atlantic States Marine Fisheries Commission 2024).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Red Drum also received a Moderate sensitivity score. This score was driven by similar attributes to our score: Population Growth Rate, Spawning Cycle, and Early Life History Requirements (Hare et al. 2016).

Exposure:

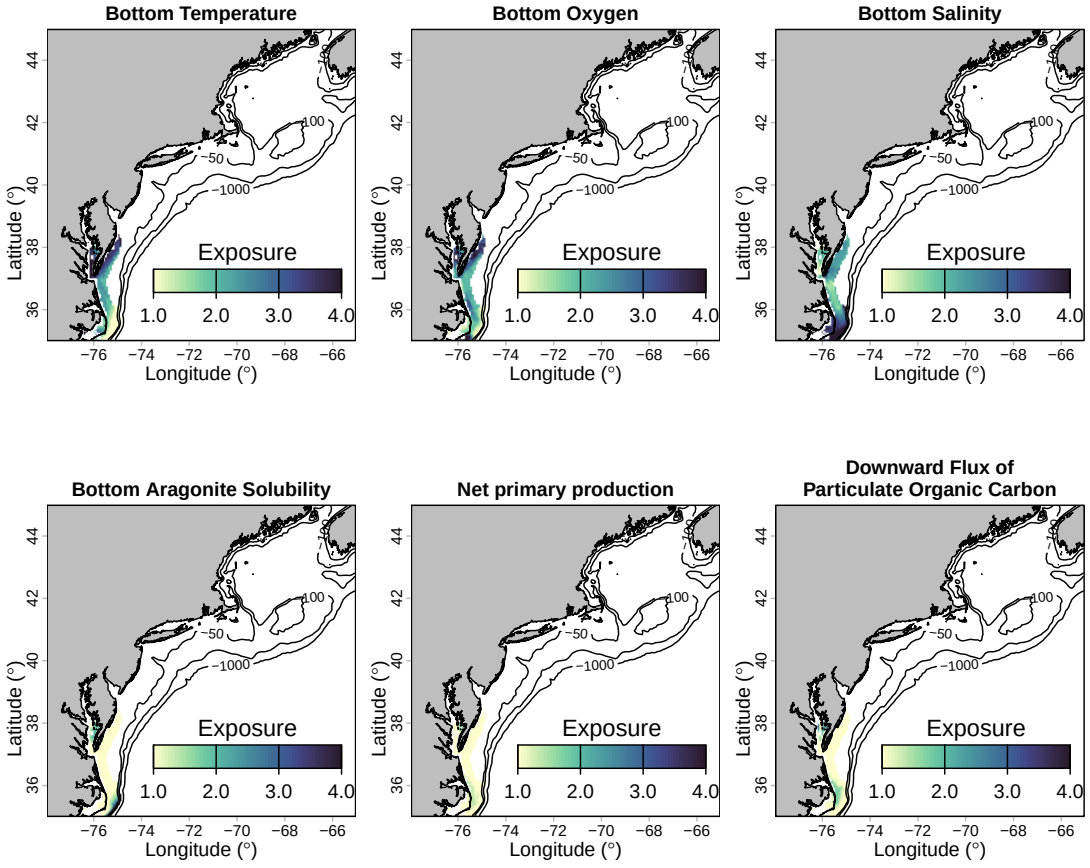


Figure 3: Maps of each Exposure Factor for Red Drum. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Red Drum was **Moderate** (1.74 +/- 0.89) (Figure 1). Exposure was greatest (High) within Chesapeake Bay, off the coast of the DelMarVa Peninsula, and off of Cape Hatteras. Exposure was

Low elsewhere in the species range (Figure 4). These patterns were driven by high bottom temperature, oxygen, and salinity exposure (Figure 3).

In the previous Assessment, Red Drum received a Very High score, driven by sea surface temperature, air temperature, and ocean acidification (Hare et al. 2016).

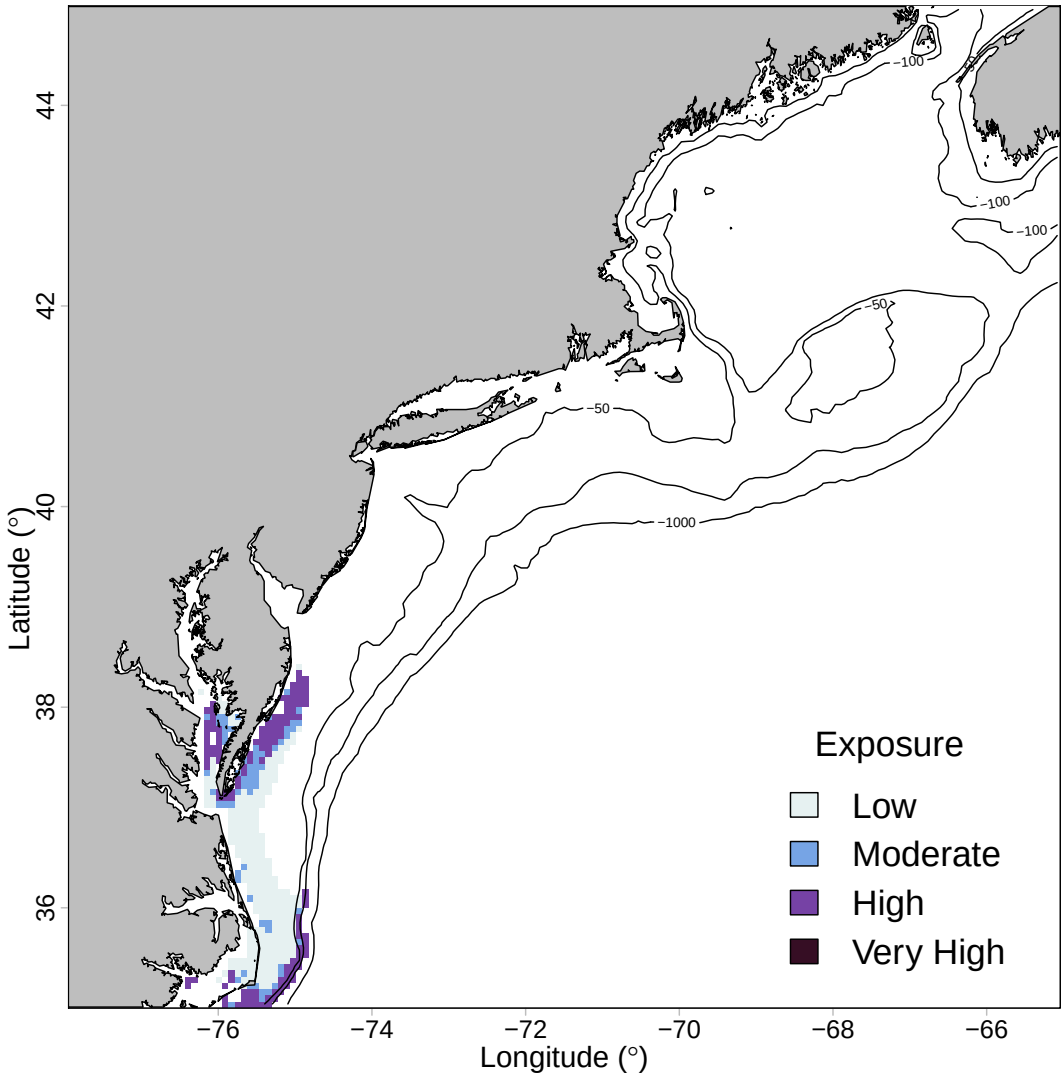


Figure 4: Total Exposure on the Northeast Shelf for Red Drum. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

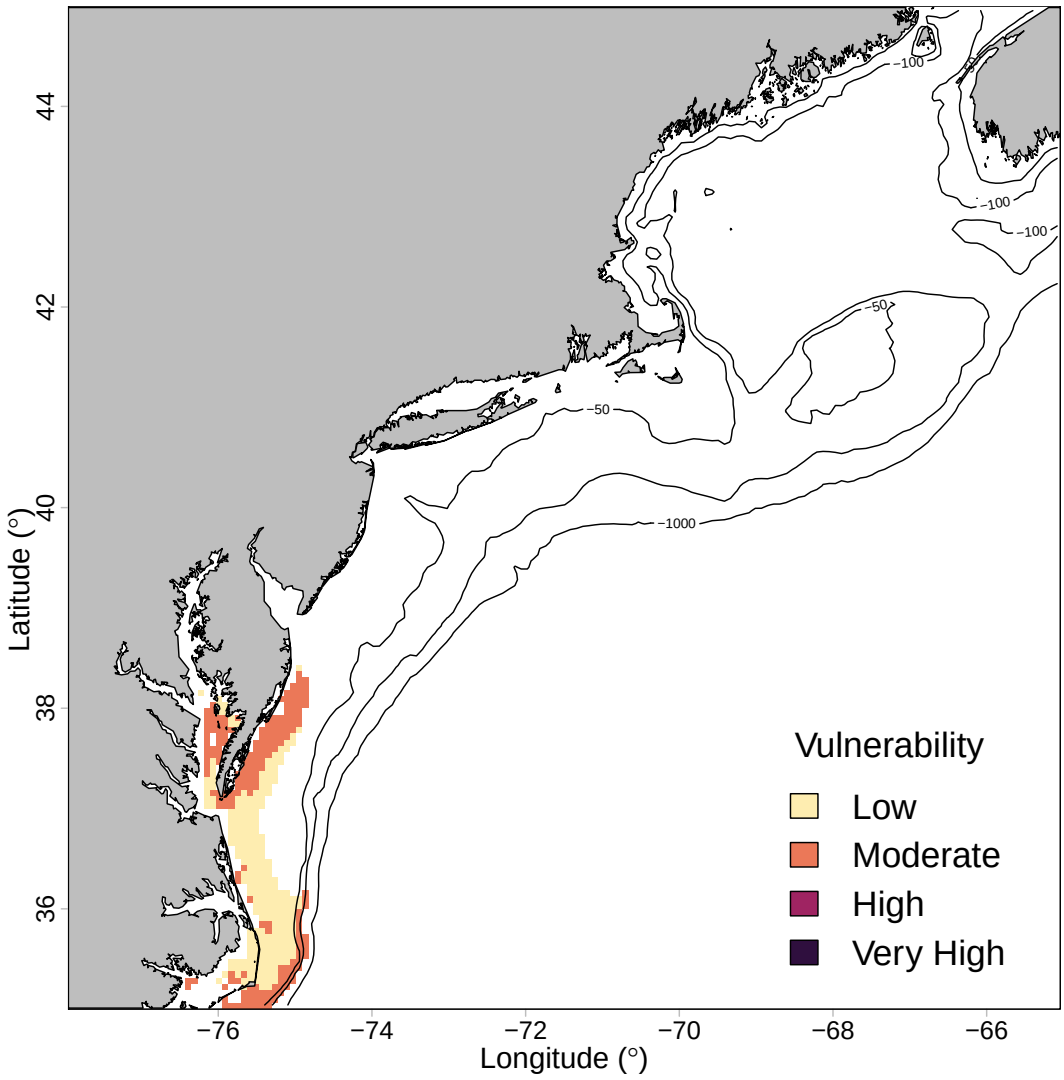


Figure 5: Vulnerability on the Northeast Shelf for Red Drum. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean overall vulnerability rank for Red Drum was **Low** (3.49 +/- 1.77) (Figure 1). Vulnerability was Moderate within Chesapeake Bay, off the coast of Virginia and Maryland, and off Cape Hatteras, but Low elsewhere within the species range (Figure 5).

In the previous Assessment, Red Drum received a High Vulnerability score due to a higher Exposure score than our assessment (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The red drum ensemble model received a mean model confidence score of 2 (+/- 0.7), indicating moderate confidence in the final model. The final model represented known distributions well, but the low sample size drove down scores. The ensemble model AUC score was 0.91, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

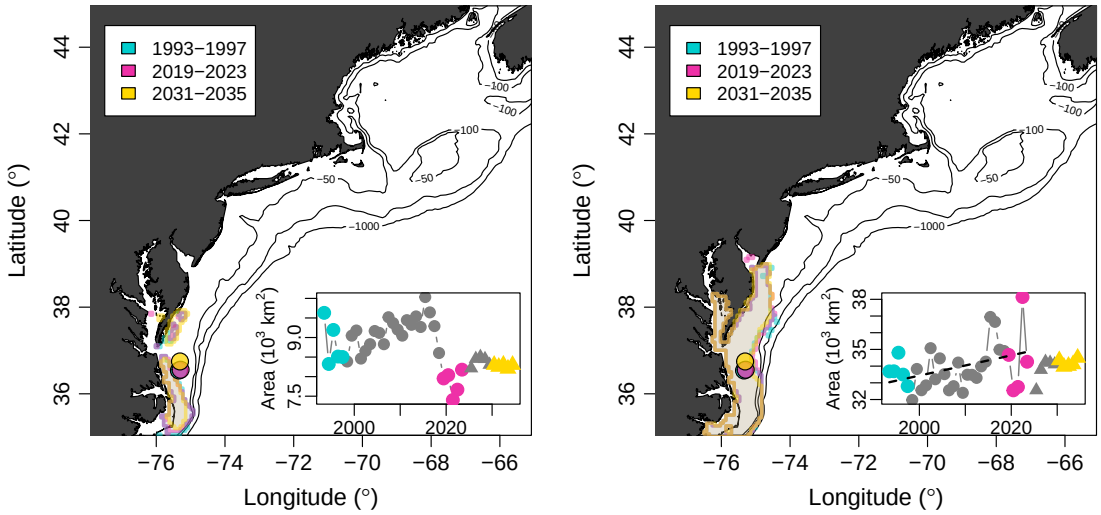


Figure 6: Distribution shifts on the Northeast Shelf for Red Drum. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Red drum ranges significantly expanded between 1993 and 2023 due to a northward shift in the northern edge of the distribution. Core habitat area remained relatively stable over the same time period (Figure 6). The weighted centroid shifted less than 5 km east-northeast.

Under near future climate conditions, red drum distributions are predicted to remain similar to current distributions (Figure 6). The weighted centroid is predicted to continue to shift northward by another 21 km.

Red drum received a Moderate Distributional Vulnerability Rank in the previous assessment due to long seasonal migrations (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	9,004	33,830	3.12	1.36
2019-2023	4.6 km ENE	8,001 (-1,002)	34,612 (+783)	3.18 (+0.06)	1.36 (0)
2031-2035	21.4 km N	8,251 (+250)	34,205 (-407)	3.18 (0)	1.36 (0)

Directional Effect:

Experts predict that climate change will have a **neutral** effect on Red Drum. Certainty was 85%. Higher temperatures have been correlated with strong recruitment events (Goldberg, Paramore, and Scharf 2022). However, Red Drum occupy vulnerable estuarine habitats, and rely on highly vulnerable habitats such as seagrass beds and shellfish reef habitats (Guyant et al. 2022). In the previous Assessment for the Northeast Shelf, experts predicted Red Drum would be positively impacted by climate change, albeit with low certainty (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Red drum recruitment has been correlated with wind directions and sea surface temperature (Atlantic States Marine Fisheries Commission 2024). In North Carolina, the duration and magnitude of N and NE winds in August was twice as strong in high recruitment years. Elevated nearshore sea surface temperatures and coastal productivity (high chlorophyll a) were positively correlated with strong year classes (Goldberg, Paramore, and Scharf 2022). Red drum acclimated to higher temperatures had a higher standard metabolic rate, decreased critical oxygen threshold, and reduced ability to recover from exercise. Crude oil exposure (hypoxic conditions) also resulted in lower metabolic rates and reduced ability to recover from exercise, and combined with climate-induced increases in temperature, are likely to impact red drum respiratory performance and fitness (Ackerly and Esbaugh 2021). Laboratory experiments on early life stage red drum resulted in decreased survival and increased heart rate when exposed to elevated pCO^2 ; however the effects were less severe on red drum compared to similar studies on other species (Lonthair, Ern, and Esbaugh 2017).

Life History Synopsis

Red drum, or redfish, are a popular recreational fish species found in estuarine and inshore waters from Massachusetts to Florida. The stock is divided into northern and southern subunits from New Jersey to North Carolina and South Carolina to Florida, respectively (Atlantic States Marine Fisheries Commission 2024). Spawning occurs within 45-60 days between August and October and where currents can move larvae/eggs into nearby estuaries and nursery habitats (Atlantic States Marine Fisheries Commission 2024).

Optimal spawning temperatures in laboratory studies ranged between 22 - 25°C, with spawning observed from 22 - 30°C (Atlantic States Marine Fisheries Commission 2024). Eggs hatch within 24-36 hours (Peters and McMichael 1987). Larvae settle at approximately 17 days old in Florida estuaries (Rooker, Holt, and Holt 1998). Temperatures of 25°C and salinities around 30 are optimal for hatching success and larval survival. Larvae growth rates were also correlated with ocean temperatures in the Gulf of America (Atlantic States Marine Fisheries Commission 2024) (Comyns et al. 1989). Larvae in Florida consume abundant copepods, whereas juveniles consume abundance amphipods, shrimp, and mysids, switching to crabs and fish as they grow (Peters and McMichael 1987). Juveniles rely on structural nursery habitats such as seagrasses for protection from predators. Juveniles are most abundant in estuarine waters and inlets, while fish older than age four inhabit deeper waters (Rooker, Holt, and Holt 1998). Adults in the Gulf of America (TX) consume white shrimp, gulf menhaden, and blue crab. Dominant prey species shift seasonally (shrimp in fall, menhaden in spring) while some prey species are consistent throughout the year (crab) (Scharf and Schlicht 2000). Age at maturity in red drum off North Carolina is 3-4 years, with a maximum age of 62 years (Atlantic States Marine Fisheries Commission 2024). Red drum move out of estuaries into coastal waters as they mature. Seasonal movements with seasonal site fidelity have been observed in Georgia but some populations (particularly in the southern stock) are resident year round. Seasonal migrations between Virginia and North Carolina have been observed between May and September (Atlantic States Marine Fisheries Commission 2024). The northern stock is managed by the Atlantic States Marine Fisheries Commission. The northern stock is not overfished and overfishing is not occurring, however the southern stock is overfished and overfishing is currently occurring (Atlantic States Marine Fisheries Commission 2024).

Supplemental Scores

Exposure Timeseries

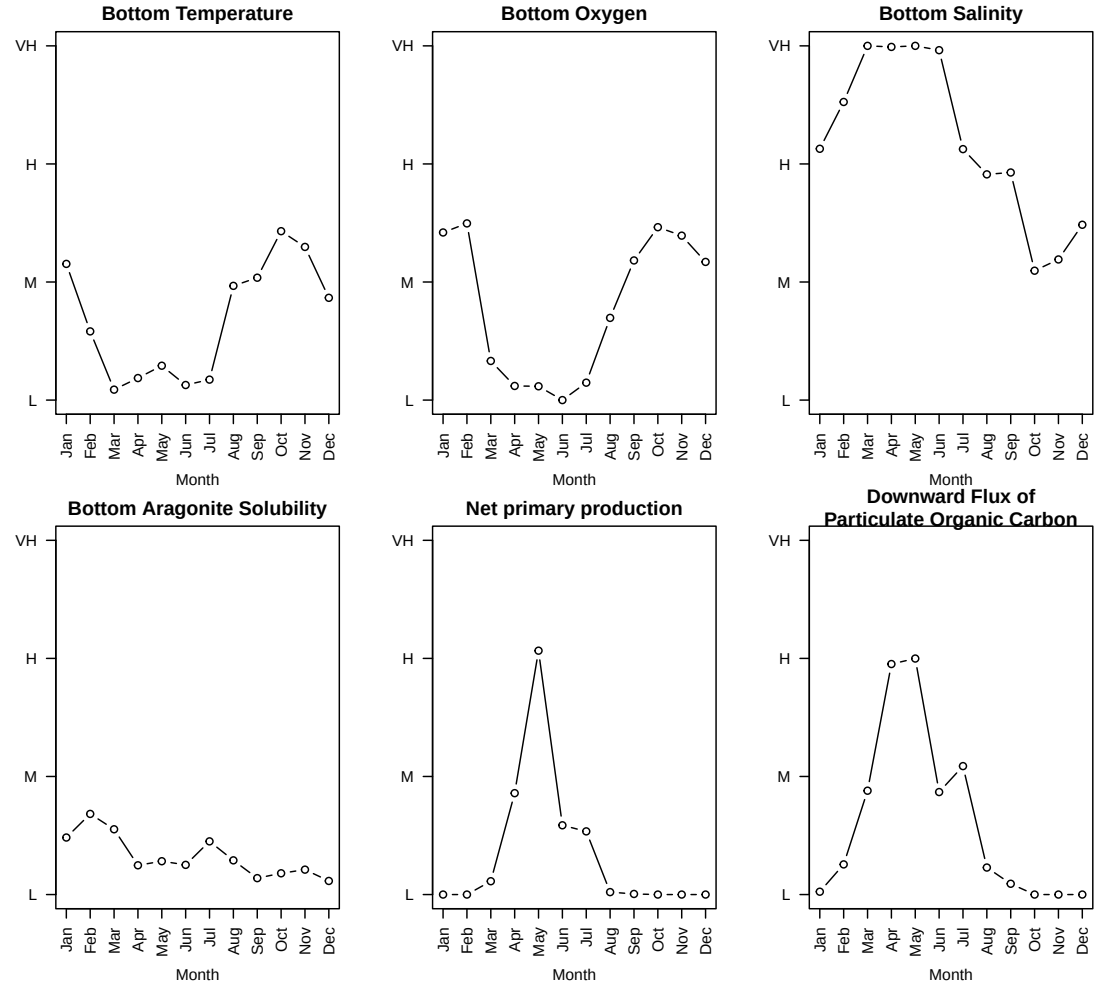


Figure 7: Monthly Timeseries of each Exposure Factor for Red Drum. Exposure was calculated between 2014-2023 and 2025-2035

Throughout the year, exposure ranged from Low to High (Figure 8). Exposure was High in May due to coincident increases in bottom salinity, net primary production, and downward particulate organic carbon flux (Figure 7).

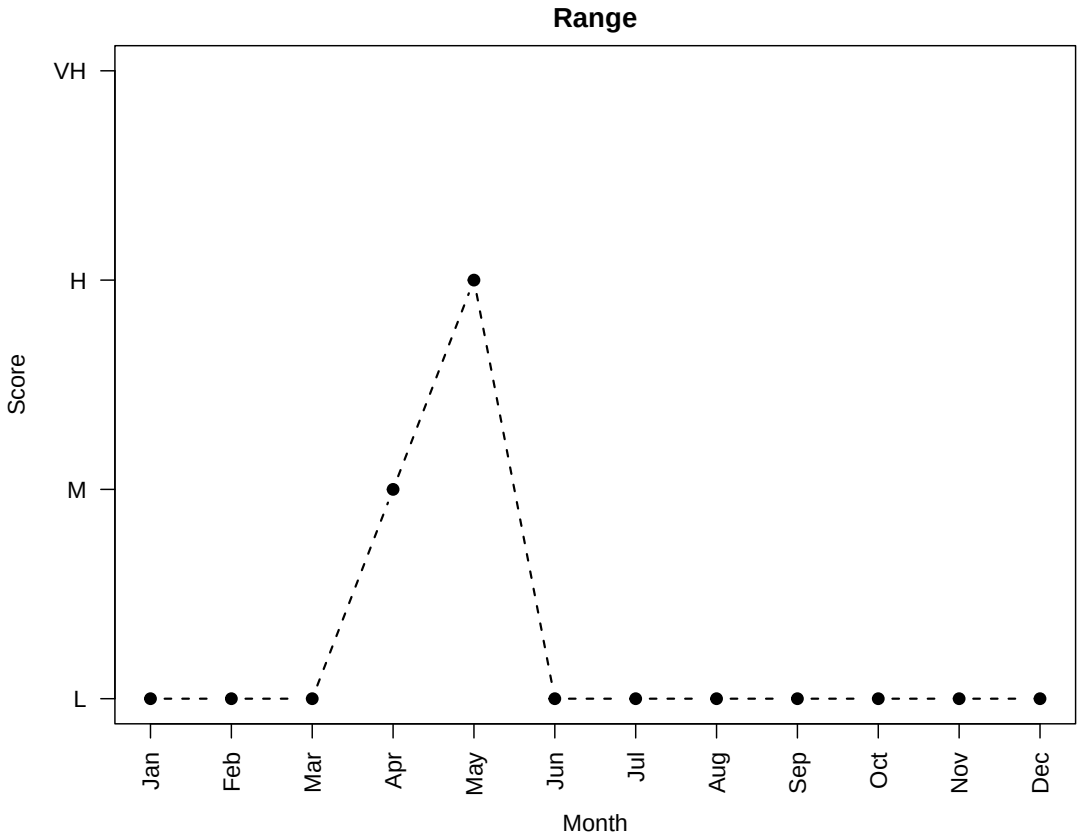


Figure 8: Timeseries of Total Exposure for Red Drum

Exposure due to Significant Factors

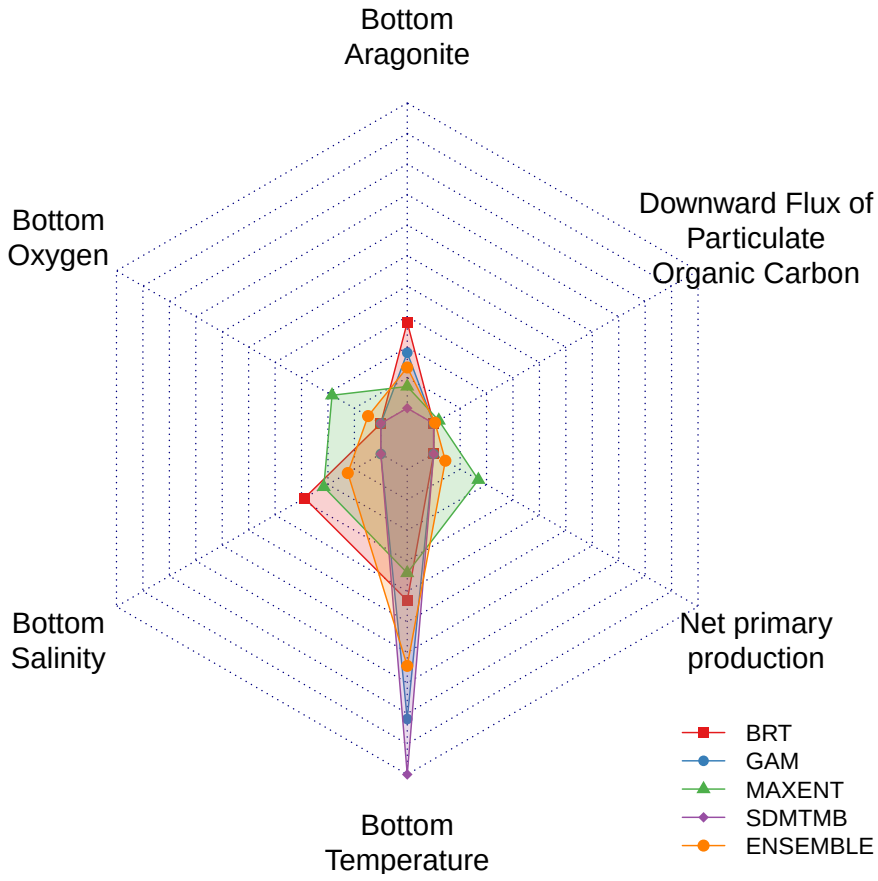


Figure 9: Dynamic Factors driving Red Drum distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, salinity, and aragonite solubility were the most important factors to Red Drum distributions (Figure 9). Estuaries are important nursery habitats and adults also utilize shellfish reef habitats (Rooker, Holt, and Holt 1998) (Guyant et al. 2022). When only these factors were used to estimate exposure, exposure decreased and was only high at the mouth of Chesapeake Bay and near the continental shelf break off of Cape Hatteras (Figure 11; Figure 10).

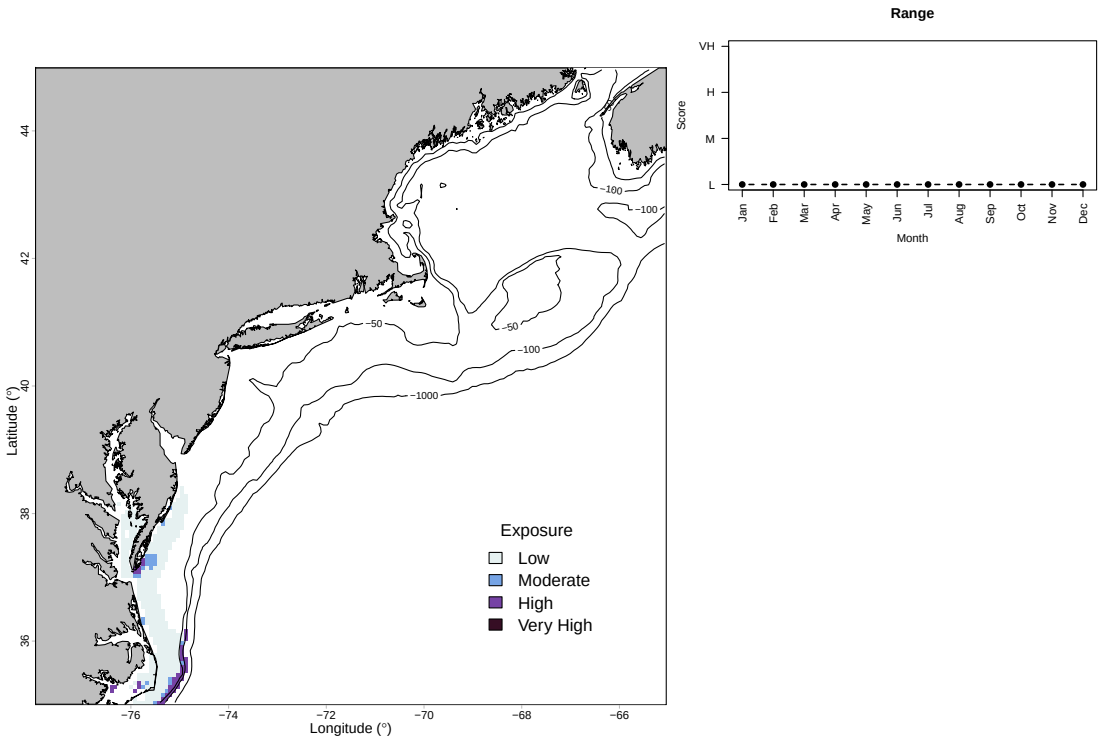
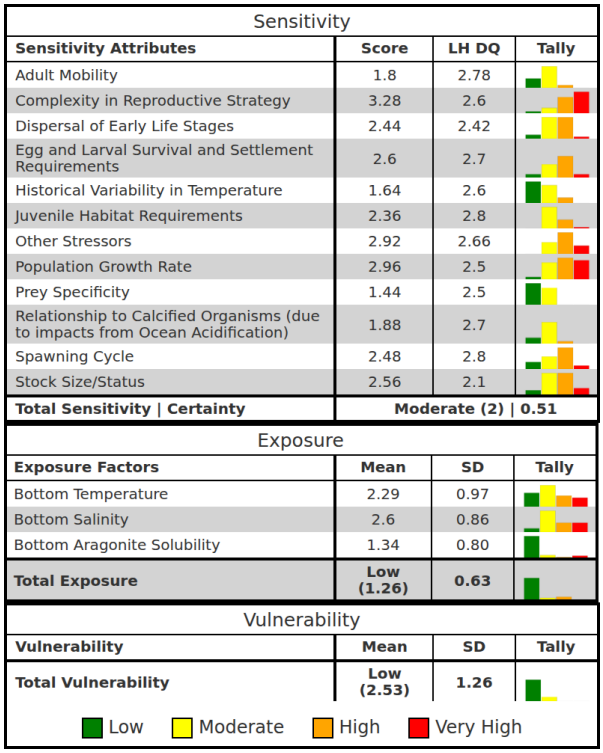


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Red Drum distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.8	2.78	
Complexity in Reproductive Strategy	3.28	2.6	
Dispersal of Early Life Stages	2.44	2.42	
Egg and Larval Survival and Settlement Requirements	2.6	2.7	
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Spawning Cycle	2.48	2.8	
Stock Size/Status	2.56	2.1	
Total Sensitivity | Certainty	**Moderate (2) | 0.51**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.29	0.97	
Bottom Salinity	2.6	0.86	
Bottom Aragonite Solubility	1.34	0.80	
Total Exposure	**Low (1.26)**	**0.63**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.53)**	**1.26**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Red Drum, but with only significant Exposure factors and the resulting Vulnerability Scores.

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